

Practice on Drafting Solutions

Create a solution template for series tests (if you still cannot write a full solution on the spot).

- How to formally verify necessary **hypotheses** (such as a function is decreasing or continuous) before applying a specific series test?
- You may split into few **cases**, e.g., the cases where an alternating series converges absolutely, converges conditionally, or diverges.
- List down some steps that are needed to be **written rigorously** (or list down what you should not write), e.g., ALWAYS write $\int_1^\infty f(x) dx = \lim_{c \rightarrow \infty} \int_1^c f(x) dx$ before evaluating improper integral; $\lim_{n \rightarrow \infty} \frac{1}{n} = \frac{1}{\infty} = 0$ is NOT acceptable.
- Include **examples** from lecture notes or tutorials.

In case you do not know what to do, try to answer the following questions.

- (1) How to justify $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n+3}}$ converges conditionally?
- (2) How to justify $\sum_{n=1}^{\infty} (-1)^n$ diverges? What about $\sum_{n=1}^{\infty} (-1)^n n$ and $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n+1}$?
- (3) What to do if the series contains trigonometric / inverse trigonometric / hyperbolic trigonometric functions? Are they sharing the same method?
- (4) How to find a candidate to be compared in Direct / Limit Comparison Test?
- (5) How to find the radius of convergence for a power series? How many cases and how to write a solution for each case?
- (6) What is the similarity between the series $\sum_{n=1}^{\infty} \sin \frac{1}{n}$ and $\sum_{n=1}^{\infty} \tan \frac{1}{n}$? Can you give other similar examples (Maclaurin series might help)?